

Design Research Practice

Design 303 | Class 10

Learning Outcomes

By the end of this week, you should be able to:

1. Understand how to use your blogs to extract and synthesize key learnings in preparation for the final presentation in week 12.
2. Identify key skills and knowledge that you have developed through your experiments.
3. Identify areas (materials, technologies, broader concerns) you would like to explore beyond this course.

Week 10 Agenda

Topic / Activity	Time
Group digest	2:05
Project exemplar: Reflections on a practice-led PhD (Leo)	2:20
Group activity: Reflecting on your research journey	2:45
Break	3:20
Overview of Final presentation and rubric	3:30
Reflections on reflections: Blog review guide (on Canvas, in 'Modules')	4:00
Break	4:20
Pair activity: Blog peer review	4:30
Studio session	4:30

Group digest – Group 15, 16

Group 15:

Group 16: <https://canva.link/8h498khygdsenk1>

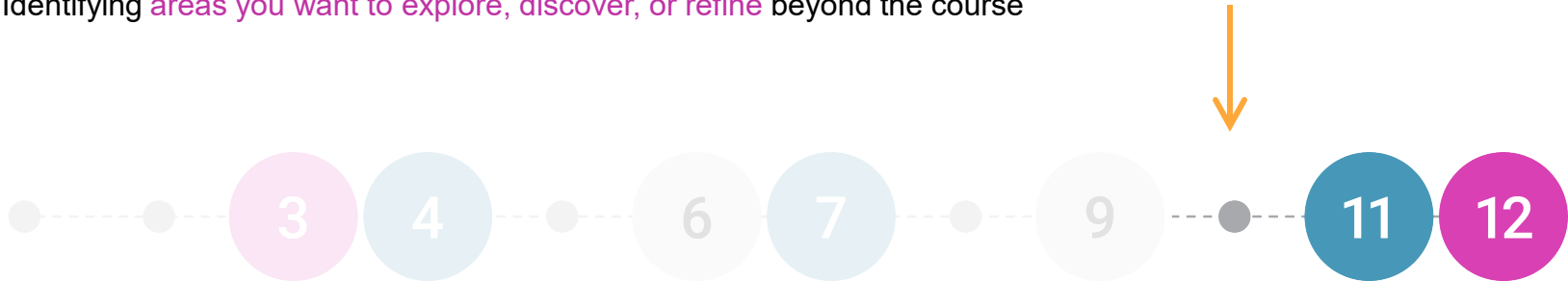
Progress Check

Where are we? Where are we heading?

You are here...

This week is about:

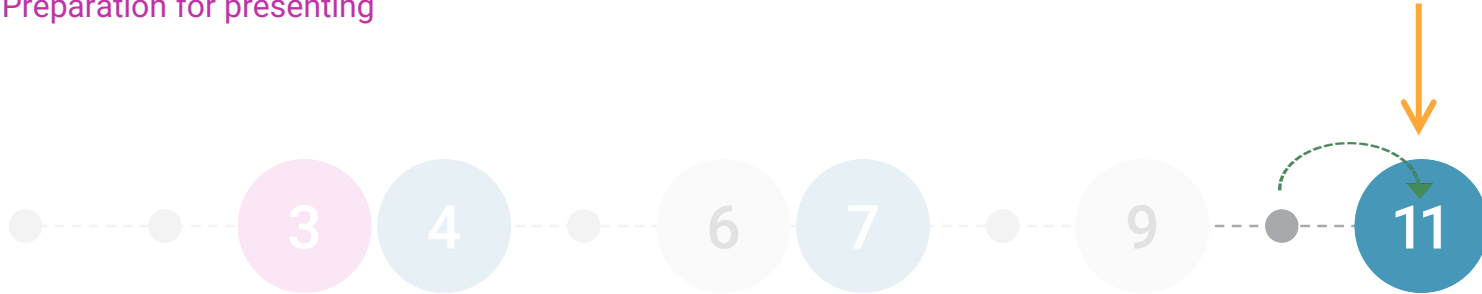
- **Reflections on reflections** – reviewing your blog to extract key learnings and form a narrative to help you prepare for the final presentation in week 12
- Identifying the **skills and proficiencies, or knowledge** of materials, technologies, or disciplines that you have developed through your experiments
- Identifying **areas you want to explore, discover, or refine** beyond the course



Class in Week 11

In week 11, the focus will be on:

- Considering key learnings from your prototyping experiments **in relation to your DES300 proposal and future directions**
- Discussing how the **skills you've developed will support your Capstone project**, as well as the **gaps or challenges** you expect to face
- **Preparation for presenting**

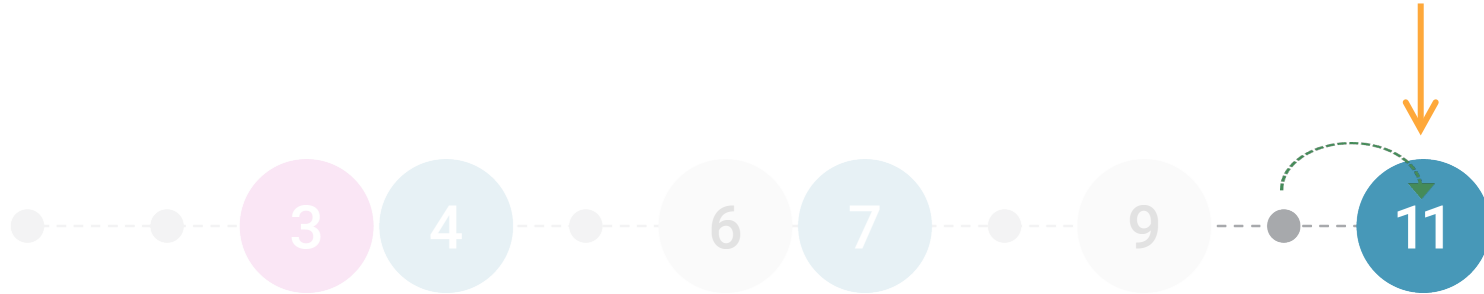


Blog update 3 is in Week 11

This is where we will assess your progress and the **outcomes of your crits!** It's worth **15%** of your overall grade.

Due Thursday 28th May by 9 am

Blog update 3



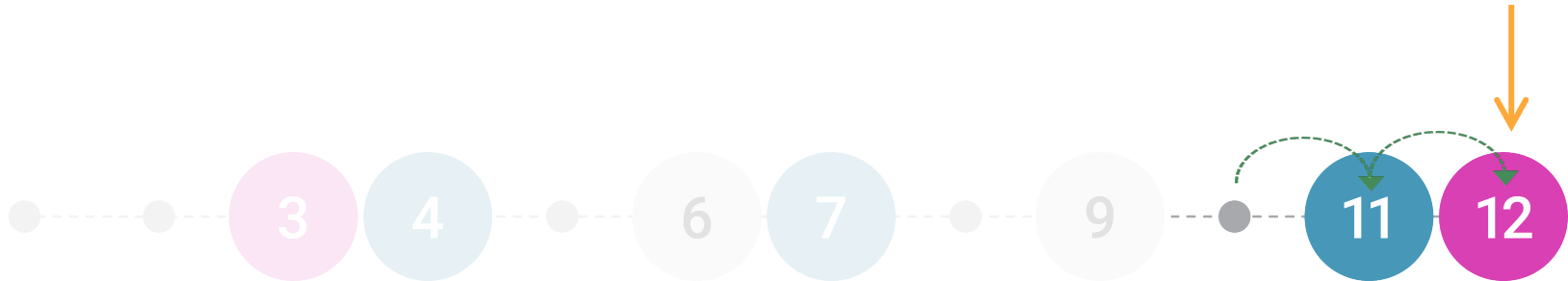
Final presentation is in week 12

The final presentation and peer feedback is worth 30% of your overall grade. You'll present a final reflective presentation showcasing your prototyping experiments, articulating connections with your DES300 proposal and potential future directions in 304.

Slides + other accompanying media Due Thursday 4th June by 2pm

Peer feedback and reflection on Feedback fruits Due Monday 8th June by 8pm

Final presentation



Project Exemplar

Reflections on a practice-led PhD

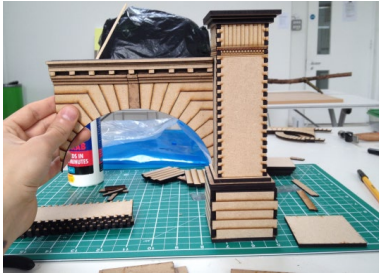
Starting Points

Postgraduate studies:

- Maîtrise, Cinema and Audiovisual Studies
- MSc Theoretical Psychoanalytic Studies
- MA Moving Image and Sound

Technologies/Materials:

- Miniature set making
- Experimental filmmaking
- Max MSP (visual programming language)





Practice-led PhD at the University of Auckland

- 4 years (2019–2023)
- Written thesis (60,000 words)
- Creative practice component (exhibition of 7 artworks)
- Supervisors: Dr Lucille Holmes & Dr Simon Ingram
- University of Auckland Doctoral Scholarship

Abstract

The work of Jacques Lacan has been influential in art and culture, yet it has largely remained absent from the theory and practice of computational arts. This research aims to remedy the gap by investigating autonomous systems in art through the lens of Lacanian psychoanalysis and its prior applications in related areas of artistic and cultural criticism. The specific theoretical constructs I employ coalesce around the subject's relationship to the machinery of symbolisation and the fundamental incompleteness of the symbolic order. By linking this theory to Alan Turing's foundational notions of computation and artificial intelligence, I show that psychoanalysis offers a pertinent body of knowledge to approach underlying themes in the field of generative computer art. In written conceptualisation and artistic practice, the thesis works towards a critical framework that considers psychoanalysis as central to understanding unique creative possibilities afforded by contemporary computing technologies.

As a basis for this framework, I re-examine the aesthetics of the uncanny to incorporate the figure of the black box, proposing a shift from the figurative and indexical to the obscurity of computation as means to evoke the unsymbolizable real. The limits of the symbolic are at stake in Turing's imitation game which poses a question about the exchangeability of human and machine, a question I relate to the dilemmas of existence and desire in obsessional neurosis by formulating simulations of obsession in art. Insofar as Turing conceived of computation by eradicating the human subject from language, psychosis is implicit as a structure in which the very accession to the symbolic is at stake, requiring a radical mode of invention to institute the subject within a signifying system. In contemporary Lacanian thought, the psychotic's tailored solution bears on the universal necessity of the symptom as an element that holds the subject together, and on the aim of psychoanalysis to provide a means of reworking the symptom. The thesis ultimately argues for the utility of computation as a material to model the (re)configuration of symptoms.

Abstract

The work of Jacques Lacan has been influential in art and culture, yet it has largely remained absent from the theory and practice of computational arts. This research aims to remedy the gap by **investigating autonomous systems in art through the lens of Lacanian psychoanalysis** and its prior applications in related areas of artistic and cultural criticism. The specific theoretical constructs I employ coalesce around the subject's relationship to the machinery of symbolisation and the fundamental incompleteness of the symbolic order. By **linking this theory to Alan Turing's foundational notions of computation and artificial intelligence**, I show that psychoanalysis offers a pertinent body of knowledge to approach underlying themes in the field of generative computer art. In written conceptualisation and artistic practice, the thesis works towards a critical framework that considers psychoanalysis as central to **understanding unique creative possibilities afforded by contemporary computing technologies**.

As a basis for this framework, I **re-examine the aesthetics of the uncanny to incorporate the figure of the black box**, proposing a shift from the figurative and indexical to the obscurity of computation as means to evoke the unsymbolizable real. The limits of the symbolic are at stake in Turing's imitation game which poses a question about the exchangeability of human and machine, a question I relate to the **dilemmas of existence and desire in obsessional neurosis** by formulating simulations of obsession in art. Insofar as Turing conceived of computation by eradicating the human subject from language, **psychosis** is implicit as a structure in which the very accession to the symbolic is at stake, requiring a radical mode of invention to institute the subject within a signifying system. In contemporary Lacanian thought, the psychotic's tailored solution bears on the universal necessity of the symptom as an element that holds the subject together, and on the aim of psychoanalysis to provide a means of reworking the symptom. The thesis ultimately argues for the **utility of computation as a material to model the (re)configuration of symptoms**.

Symptoms in Silico: Psychoanalysis and Generative Computer Art

The impulse for this research arose from my curiosity about the possibility of making connections between autonomous computer systems and the ways in which the unconscious manipulates language and visual material beyond our conscious control, such as it does in dreams, symptoms, slips of the tongue, jokes, and bungled actions.

Thus, my point of departure was an interest in creatively exploiting the conjunction of computing practices and the theory of the unconscious in psychoanalysis. Over the course of the research project my engagement with psychoanalytic ideas and their application in art and culture further refined how the investigation relates psychoanalysis and computational arts.

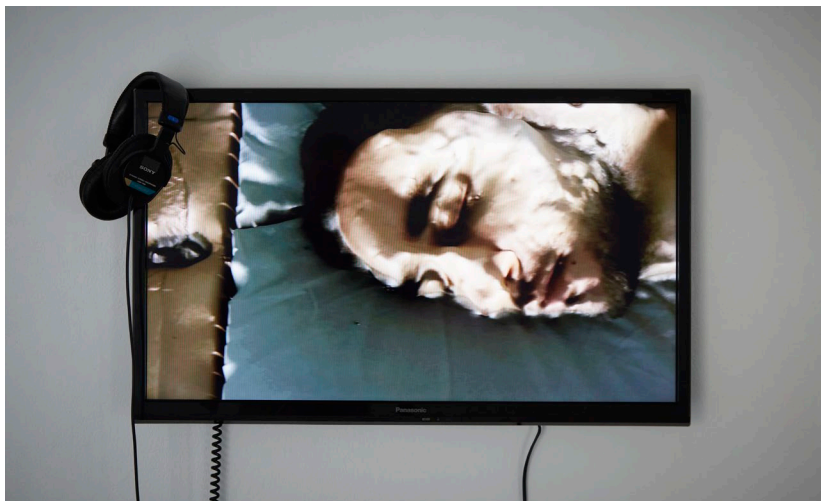
Main Contributions of the Research

- Psychoanalytic ideas applied to the theory and practice of generative computer art
- AI and the aesthetics of the uncanny (specifically the figure of the black box)
- Lacanian ideas about creativity and psychosis in relation to computing practices

Methodology

- Methodological framework grounded in psychoanalysis
- Drawing on Malcolm Quinn's 2011 essay "Insight and Rigour: A Freudo-Lacanian Approach"
- Starting from a position of not knowing, rather than leveraging an artistic identity
- "The problem teaches"

One way this is evinced is the use of different technologies, software, programming languages, computing peripherals, and other materials over the course of the research project. In making each of the seven works shown in the exhibition and presented in this thesis, I began without knowing how my practice would accommodate new conceptual elements from an evolving Lacanian 'toolkit'.



The Hole in the Mirror Machine (2020)

3D animation, sound, digital photography, machine learning text generator (fine-tuned GPT-2). 3m 35'.



The Ultimate Machine (2021)

Computer, LCD monitor, keyboard, mouse. Custom interactive Java program running on Windows.



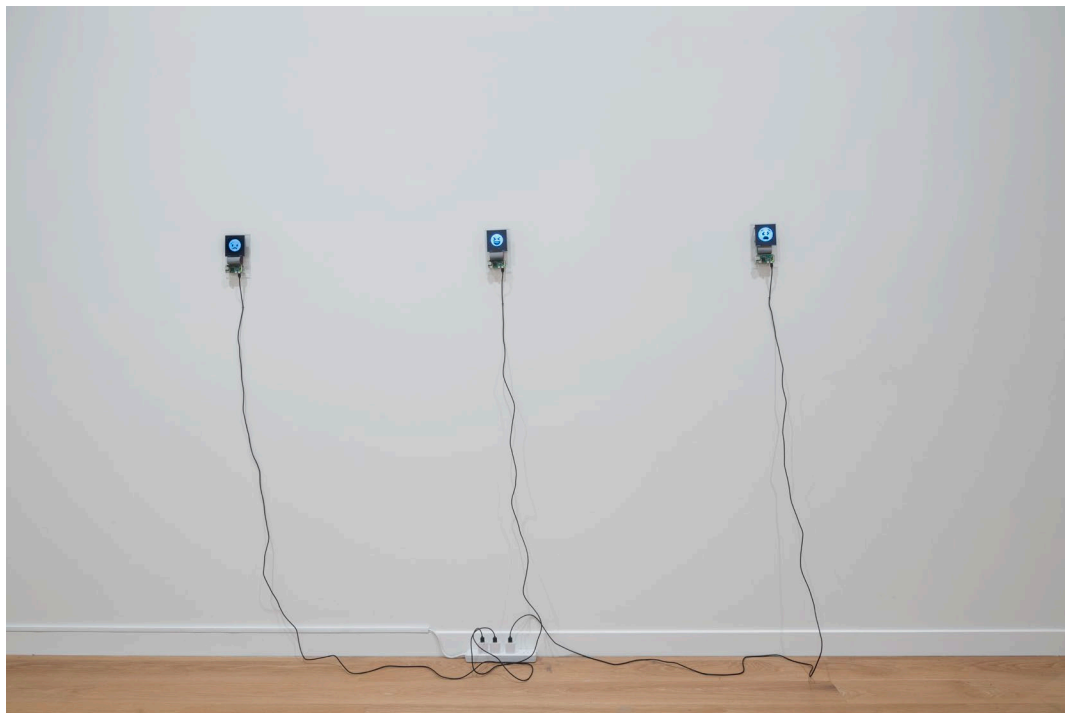
Existential Receipt Printer (2022)

Computer, thermal receipt printer, continuous paper. Custom Java program running on Windows.



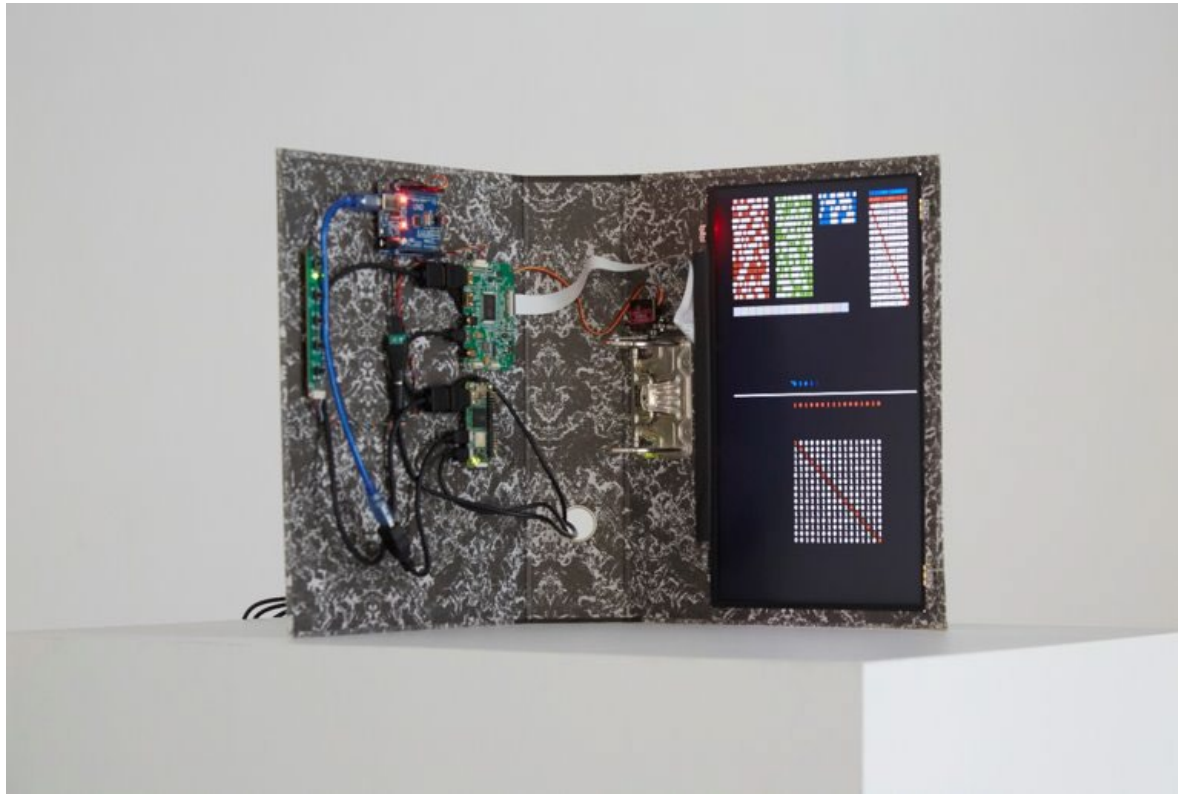
Machine Learning Living Death (2022)

Computer, LCD monitor. Outputs from a custom JavaScript program implementing a random walk through the latent space of a machine learning model (DCGAN).



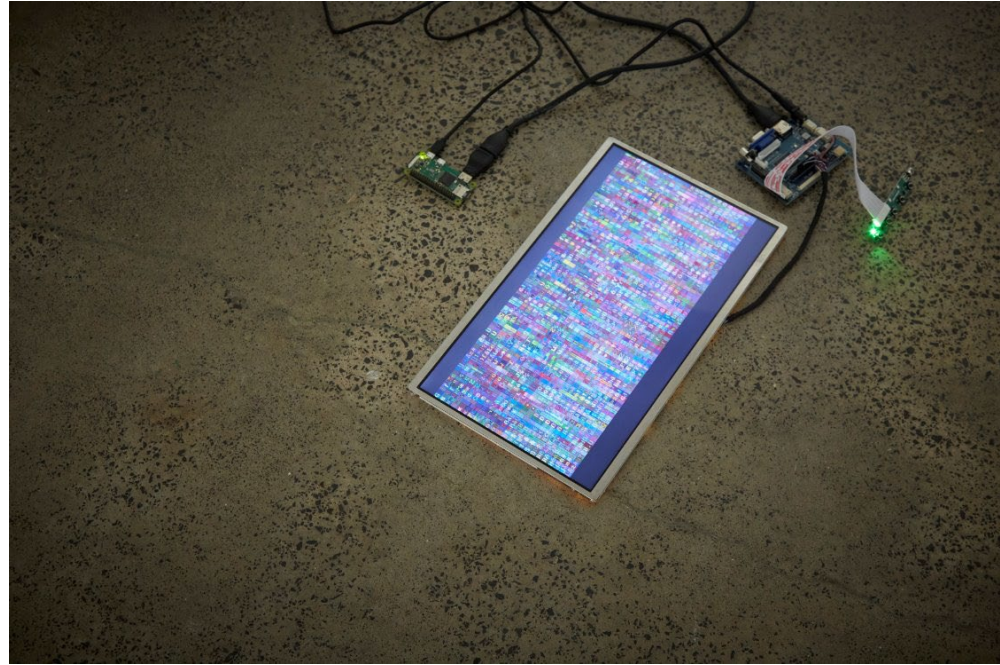
Emojis Outside Social Links (2023)

Computers, LCD screens, cables, acrylic. Outputs from a custom JavaScript program implementing a random walk through the latent space of a machine learning model (DCGAN).



Universal Bachelor Machine
Implementing Cantor's Diagonal for
the Subject to Effectuate Castration,
Even (2023)

Computer, microcontroller, lever arch file, LCD
screen, LCD controller board, servo motor. Custom
Java program running on Linux.

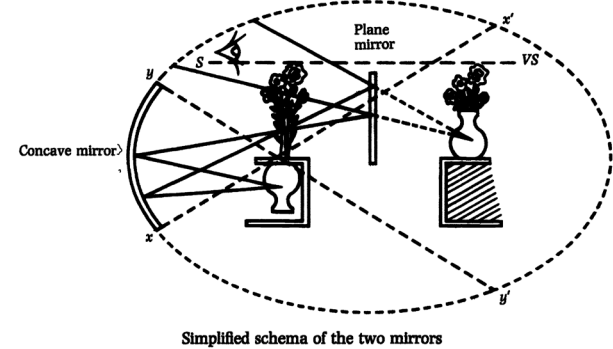


Recrasher: Virus Behaving Favourably (2023)

Computer, LCD screen, LCD controller board. Custom QBASIC program running on an MS-DOS emulator on Linux.

Challenges

- The scope and complexity of Lacanian theory
- Finding a rhythm with alternating phases of writing and creative practice



"I am thinking where I am not,
therefore I am where I am not thinking
.... I am not, where I am the plaything
of my thought; I think about what I am
where I do not think I am thinking."

Jaques Lacan, *Écrits*

Failures

- Losing focus on the project by taking on too many other things (teaching, reading groups, clinical training in psychoanalysis, conferences, other research projects)
- Waiting 1.5 years before learning a conventional, text-based programming language
- Being a 'trainspotter'

"Leo has had a busy 12 months and progress has been merely satisfactory... he has taken on too much outside of his primary (research) purpose... it's hugely important that he return to core business for the duration of his candidacy."

2021 Annual Report

Surprises & Lessons Learned

- First key creative work involved recent machine learning tools, while the final one used software from the early 1990s (QBASIC)
- A conventional approach to academic writing was far more effective than an experimental one
- Artist-researcher $\neq$ artist

... adopting the position of a researcher in relation to one's practice necessitates a shift in the structure of artistic self-identity



Experimental writing in my MA thesis

Questions for Future Research

- How might an artist construct a personal computing environment to model symptomatic reconfigurations?
- How might digital assistants, telemetry tools, operating systems, mobile devices, and computer networks be reconfigured and reassigned symptomatically?
- How might AI-enabled robotic companions become partners to a symptom?





Questions

Reflecting on Your Research Journey

40 mins

In groups (3-5 people), take turns to reflect on and share your own research journey over the past 10 weeks.

- Where did you **begin**? (*e.g. a technology, social issue*)
- What were some **defining moments**? (*e.g. discoveries, pivots*)
- When did you encounter **challenges**?
- What were your **failures**?
- What **surprised** you? What did you **learn about yourself**?
- What do you think is your **strongest** direction? What alternatives do you see?
- Where might you be heading **next**?

Break!

Final presentation overview and rubric

Your final presentation – reflective 12-week journey

What we are looking for:

1. **Connection to DESIGN 300 proposals:** how well your experiments connect to your DESIGN 300 proposal
2. **Skill development:** what you actually learned by making things
3. **Lessons learned and future plans:** where this work is taking you in Capstone
4. **Slide quality:** how well the deck communicates all of the above

Your final presentation – reflective 12-week journey



2026 Semester One

Home

SET Evaluations

Announcements

Syllabus

Modules

Grades

Assignments

People

Panopto Video

Discussions

Learning Essentials

Reading List

Zoom 1.3

Course Analytics

Disability Support

Lucid (Whiteboard)

Ed Discussion

Portfolio



Account



Dashboard



Courses



Calendar



Inbox



History



Help

**Blog: Week 7 progress check**

Due 1 May at 22:00 | -/15 pts

**Blog: Week 11 progress check**

Due 28 May at 9:00 | -/15 pts

▼ In-class critique events

10% of total

**Week 6 Crit session 1**

Due 23 Apr at 14:00 | -/5 pts

**Week 9 Crit session 2**

Due 14 May at 14:00 | -/5 pts

▼ Week 3 tech demo

15% of total

**Demo presentation & file uploads**

Due 19 Mar at 22:00 | 0/10 pts

**Peer review of tech demos 2026**

Due 23 Mar at 20:00 | -/5 pts

▼ Week 12 final presentation

30% of total

**Final presentation & file uploads**

Due 4 Jun at 14:00 | -/20 pts

**Peer feedback of final presentations (tutors assessed)**

Not available until 4 Jun at 14:00 | Due 8 Jun at 17:00 | -/10 pts

[Final presentation Link](#)[Peer review Link](#)

Your final presentation – reflective 12-week journey

Criteria	Ratings				Pts						
Connection to DESIGN 300 Proposal How meaningfully do the experiments connect back to the proposal, and what has the student learned about the issue along the way?	5 to >3.95 Pts 'A' Range The work is grounded in a specific local issue from the DESIGN 300 proposal, with explicit links between experiments and parts of the proposal. Multiple design opportunities are surfaced and connected to DESIGN 303 experiments. Gaps or pivots are openly identified, reasoned about causally, and shown to reframe understanding of the issue, the brief, and possible Capstone directions.	3.95 to >3.2 Pts 'B' Range Clear links between experiments and the proposal. Design opportunities are identified, though not all are connected back to experiments. A gap or pivot is identified and described, with some causal reasoning, but its impact on understanding the issue and Capstone direction is not fully drawn out.	3.2 to >2.45 Pts 'C' Range Some connection between experiments and the proposal is shown. Design opportunities are mentioned in passing. A gap or pivot may be named but is not analysed or causally explained.	2.45 to >0 Pts 'D' Range Lacks connection to the DESIGN 300 proposal or a specific local issue. Design opportunities are not identified. Gaps or pivots between the proposal and the experiments and prototypes are not surfaced.	5 pts	Lessons Learned & Future Plans How clearly has the student mapped where they're heading in Capstone, and how thoughtfully have they considered who they are in relation to the work?	5 to >3.95 Pts 'A' Range Key takeaways are clearly articulated. A strongest Capstone direction is identified with specific refinements, alongside at least two genuinely distinct alternative directions for Week 1 of Capstone. Near-term and longer-term visions are both set out. Positionality is acknowledged, analysed for how it shaped specific decisions, and translated into a changed way of operating as a designer and researcher. Specific things to test, build, or clarify before Capstone are named.	3.95 to >3.2 Pts 'B' Range Takeaways are clear and a Capstone direction is named with some refinement. Alternative directions are mentioned in passing. Positionality is acknowledged and its effect on the work is partly analysed, but the implications for how to operate differently are not fully drawn out. Some next things to test or clarify are named.	3.2 to >2.45 Pts 'C' Range Some takeaways are offered. A Capstone direction is mentioned, though alternatives are absent or undifferentiated. Positionality is mentioned but its effect on decisions or practice is not unpacked. Next steps before Capstone are general rather than specific.	2.45 to >0 Pts 'D' Range Takeaways are vague or missing. No clear Capstone direction or alternatives. Positionality is not addressed. Nothing concrete is identified to test, build, or clarify before Capstone.	5 pts
Skill Development & Application What concrete growth, failure, and learning has the student evidenced through their prototyping?	5 to >3.95 Pts 'A' Range Specific skills, tools, and techniques are named, with clear evidence of how feedback was recorded. Growth is evidenced through before-and-after comparison of prototypes. At least one failure is	3.95 to >3.2 Pts 'B' Range Specific skills are named with some account of recorded feedback. Growth is evidenced and direction changes are acknowledged. A failure is acknowledged, with some lesson drawn — but the lesson is not specifically	3.2 to >2.45 Pts 'C' Range Some skills or tools are named. Feedback may be mentioned in passing. Growth, direction changes, or failure may be discussed in general terms but are not concretely evidenced or	2.45 to >0 Pts 'D' Range Skills, tools, and techniques are not named concretely. No record of feedback. No clear evidence of growth, direction change, or honest discussion of failure.	5 pts	Slide presentation quality How well does the deck itself communicate the work through evidence, design, structure and clarity?	5 to >3.95 Pts 'A' Range Slides are readable and free of obvious errors. Visuals are the student's own prototype work — photos, sketches, screenshots — and serve as direct evidence for specific claims. The deck is consistent and uncluttered, and feels deliberately designed with considered hierarchy and pacing.	3.95 to >3.2 Pts 'B' Range Slides are readable and visuals are mostly the student's own work with relevance. Most visuals serve as evidence for claims. The deck is generally consistent, though hierarchy and pacing could be more deliberately considered.	3.2 to >2.45 Pts 'C' Range Slides are readable but visuals include stock imagery or are not clearly tied to claims as evidence. Consistency in typography, spacing, and density across slides could be tightened.	2.45 to >0 Pts 'D' Range Slides have readability issues, typos, or broken layouts. Visuals are missing, stock-based, or do not function as evidence. The deck reads as assembled rather than designed.	5 pts

Your final presentation – reflective 12-week journey

Peer feedback of final presentation (tutor assessed)					
Criteria	Ratings				Pts
Reflection Quality	50 to >39.5 Pts A Clearly identifies the main feedback received and thoughtfully evaluates which comments were useful, challenging, or less relevant, with reasons. Insightfully explains what the feedback reveals about the project, presentation, or design approach, and identifies clear learning. Proposes specific, realistic, and relevant next steps for improving the project, presentation, or design process.	39.5 to >32.5 Pts B Identifies the main feedback and offers some evaluation of its usefulness. Explains what the feedback suggests about the work and identifies some learning. Proposes relevant next steps, though they may be somewhat general.	32.5 to >24.5 Pts C Identifies some feedback, but evaluation is limited or unclear. Shows limited analysis and only partial learning. Suggests vague or limited next steps.	24.5 to >0 Pts D Gives little or no clear account of the feedback, or accepts/rejects it without explanation. Reflection is mostly descriptive, with little or no analysis or learning. No meaningful action plan, or actions are unclear or unrelated.	50 pts
Feedback Quality	50 to >39.5 Pts A Feedback is clear, precise, and grounded in specific observations from the presentation and other media. Feedback clearly helps the presenter improve, with useful suggestions and clear next steps. Feedback is strongly related to the brief or presentation aims and is communicated respectfully and professionally.	39.5 to >32.5 Pts B Feedback includes some specific observations and is mostly clear. Feedback is constructive and includes at least one helpful suggestion. Feedback is mostly relevant and respectful.	32.5 to >24.5 Pts C Feedback is somewhat general, with limited reference to the actual presentation. Feedback shows some constructive intent, but suggestions are limited or unclear. Feedback is partly relevant or uneven in tone.	24.5 to >0 Pts D Feedback is vague, generic, or not clearly based on what was presented. Feedback is superficial, unhelpful, or gives no direction for improvement. Feedback is irrelevant, careless, or communicated in an inappropriate tone.	50 pts
Total points: 100					

10%

Your final presentation – reflective 12-week journey

Purpose: Developing the capacity to learn from your own work

Experiments, crits, documentation all support you to practice the loop of making: trying, failing, examining, extracting lessons, changing approaches, trying again.

– The rhythm of self-directed 304 capstone

The presentation should articulate what you've learnt on this journey, and where this might then direct you next into 304.

~~Did I prototype enough things? Can I show what I wanted to learn and why?~~

Your final presentation – 1. Connection to Design 300 proposals

How well your experiments connect to your DESIGN 300 proposal:

Key questions to consider:

What specific local issue from your DESIGN 300 proposal does your work address?

How do your experiments map onto a part of that proposal?

What different design opportunities does your DESIGN 300 proposal reveal, and how have your DESIGN 303 experiments helped you recognise them?

Where did a gap, mismatch, or pivot emerge between what you planned and what actually happened?

Why did that gap or pivot happen?

How did the pivot reshape your understanding of the issue, the brief, or your possible Capstone directions?

Your final presentation – 2. Skill development

What you actually learned by making things:

Key questions to consider:

What specific skills, tools, or techniques did you engage with through prototyping?

How did you record feedback?

How has your work changed from your earliest prototype to your latest? When did you have to change your direction, why?

What failed, broke, or had to be abandoned along the way?

What did that failure teach you?

Which specific skill do you now need to develop, and why?

Your final presentation – 3. Lessons learned and future plans

Where this work is taking you in Capstone:

Key questions to consider:

What key takeaways have emerged from your experiments and research?

Which aspect of your experiments now feels like your strongest Capstone direction, and what would you refine or explore further?

Where do you see this work in the near (i.e. in a month?) or distant future (i.e. mid of capstone)?

What are at least two other possible directions you could present in Week 1 of DESIGN 304, and how are they different from your strongest direction?

How has your positionality shaped specific decisions in this work?

In what ways will you operate differently as a designer and a researcher?

What would you do differently next time, and what does this suggest you should test, build, or clarify before DESIGN 304 begins?

Your final presentation – 4. Slide quality

How well the deck communicates all of the above:

Key questions to consider:

How readable are your slides and have you proofread them?

Whose work appears in your visuals yours, or stock imagery?

What specific claim does each of your visuals serve as evidence for?

How consistent is your typography, spacing, and density across the deck?

What does deliberate design look like in your deck. What hierarchy and pacing decisions did you make?

Do you need to submit supplemental scripts, videos or other information?

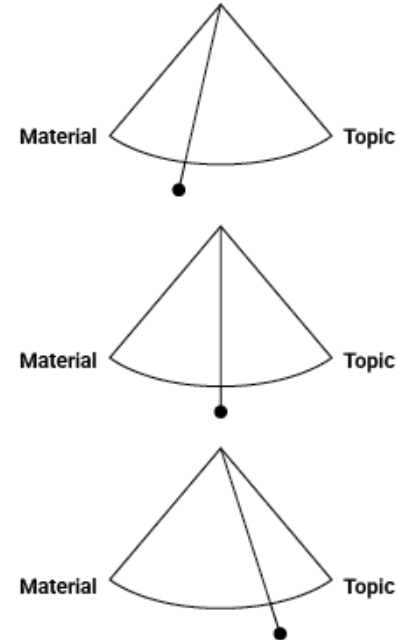
Reflections on Reflections

Blog review guide (on Canvas, in 'Modules')

Reviewing Your Design Research Journey

There is no single 'correct' journey

- Some of you were more interested in technical, material, and craft experiments from the outset
- Others were more drawn to social or conceptual issues initially (responding creatively to a topic)
- Many were somewhere in-between at the beginning
- Along the way, your focus may have shifted
- You might have arrived at unexpected results



Reviewing Your Design Research Journey

Make a chronology of your experiments

- Extract your action plans from each blog post (summarise if needed)
- Evaluate your progress with each plan
- Select screenshots/images/visuals from each experiment
- Use these elements to build a more concise narrative of your journey
- Describe the different directions that came out of your experiments
- Describe your final experiment and the key factors leading to it

Experiment Progression

Suggestion: Make an overview of your posts using the IRC

A table to show both depth (each column = complete IRC) and growth (row by row evolution).

IRC Step	WK 1	WK 2	WK 3	WK 4	WK 5	WK 6	WK 7	WK 8	WK 9	WK 10
Experience										
Reflection										
Theory										
Preparation										

Example: Tim's Experiment Progression (Weeks 4–6)

<https://timbradfield.substack.com/>

IRC Step	Week 4 – Direction setting	Week 5 – Momentum	Week 6 – Reframing
Experience	Brief scoping; in-class ideation prompts; 30 quick ideas; lands on low-fi modular stall concept	Direction shifts to rain; rain-drum + shelter idea; Exp 1 (rain-drum shelter), Exp 2 (vessel size/material sound tests)	Crit week; Salone digest + crit slides; Exp 3 (perforated-container controlled droplet flow); drainage/acoustics sketches
Reflection	Resisting familiar/prestigious choice; wants physical strategic output; admits notification distraction, over-ambition	Admits over-attachment to stall concept; best insights from un-tested accidents (drainage, lid vibration); rain as "stakeholder"	Crit reframes rain as experience not problem; drainage now core not secondary; structure may matter more than the drums
Theory	Best Awards student work playful, resolved, human-centred benchmarks	Gunnera biomimicry; TikTok rain-drum critiques; Harry Bertoia Sonambient sound sculptures	Erik Vallbo <i>Rainwaves / Time Flies</i> (via Shen thesis); Aēsop Aposē collage; Hannes Peer <i>La Casa di Marmo</i>
Preparation	Manage ambition vs momentum; clarify brief choice; sketch modular stall concepts	More focused experiments; plan the crit (idea→test→observation→next); consult metal/architecture workshop staff	Semester mapped out; park structural feasibility, deepen sound; Fab Lab + metal-tech consults; DES300 research proposal

Reviewing Your Design Research Journey

Prepare slides for your presentation

Presentations should be 10–15 minutes total. In week 11, we'll focus on how your experiments might support your Capstone project — this should also be included in your presentation.

This week's guide can help you with the following elements:

- Introduction: introduce yourself, your research journey, and briefly outline what you'll cover in the presentation.
- Experiment Progression: overview of your prototyping experiments
- Key moments: explain important decisions along the way in relation to key learnings
- Highlight failures
- Use visuals, images of prototypes, diagrams, etc. as evidence and to reinforce messages
- Outline how the experiments have informed intentions for future work



Questions

Break!

Blog Peer Review

Blog Peer Review

15 mins each

In pairs, read your partner's blog (in full) and make notes:

- What are the **defining moments**?
- Where is **growth demonstrated** (skills and knowledge)?
- Are any **failures** highlighted?
- Is there anything **surprising**?
- Which is your **strongest** direction? What alternatives do you see?
- How might the experiments inform **future work**?

Then, take turns to share and discuss your peer review notes:

Independent Study/ consults

Independent Study

Things to complete this week

- **Keep making!**
- **Write your final blog post** (*if you have found the process useful, keep blogging to help you plan for your final presentation!*)
- **Your week 10 blog should include planning towards the final preparation**
- **Use your blog review to inform your final experiment and as preparation for the final presentation in week 12**

***Reminder:
Week 11 blog update is
due next week, Thursday
28th May by 9 am***

Thank you!
Mā te wā...